SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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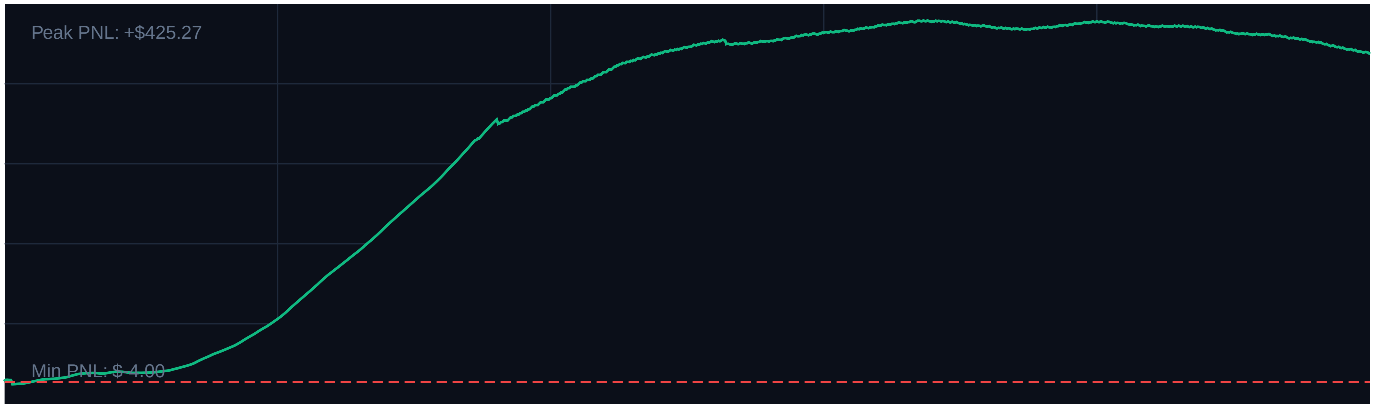
TARGET INSTRUMENT: R_75 Volatility Model

CORE PERFORMANCE METRICS

Total Settlements:	1000	**Calculated Profit Factor:**	3.16
Win Accuracy:	65.5% (655W / 345L)	**Max Peak Drawdown:**	0.4%
Cumulative PnL:	$386.15	**Avg. PnL per Trade:**	$0.39

EQUITY CURVE PROGRESSION



Initial

Trade 1000 (PnL: $386.15)

PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

SEGMENT	WIN RATE	NET PnL	REFINEMENT EFFICIENCY
Trades 1-200	87.5%	$72.77	+105% [OK]
Trades 201-400	96.0%	$260.99	+115% [OK]
Trades 401-600	61.0%	$76.70	+73% [OK]
Trades 601-800	50.0%	$13.80	+60% [OK]
Trades 801-1000	33.0%	$-38.11	+26% [OK]

SOVEREIGN NARRATIVE ADVISORY

Cognitive Strategy Proposal & Adaptive Intelligence Logs

SOVEREIGN AI: ALGORITHMIC RISK AUDIT & SYSTEM RE-CALIBRATION
Target Book: Multi-Asset Trade Settlement Book (N=1000)
Security Status: Active Risk Evaluation

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1. EXECUTIVE TELEMETRY CRITIQUE

Based on the parsed telemetry of the 1,000-trade sample, the system exhibits extreme structural anomalies that indicate high theoretical edge but severe real-world capital inefficiency.

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| Metric                  | Value           | Sovereign Risk Evaluation      |
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| Win Rate                | 65.5%           | Statistically robust (Z-Score > 3)|
| Profit Factor           | 3.16            | Institutional grade (theoretical)|
| Cumulative PnL          | $386.15         | Highly critical underperformance |
| Average PnL per Trade   | $0.386          | Sub-optimal / Commission-negative|
| Maximum Peak Drawdown   | 0.4%            | Severely under-leveraged       |
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 # Diagnostic Findings:
***   **The Micro-PnL Paradox:** While a Profit Factor of 3.16 and a 65.5% win rate are mathematically elite, a cumulative return of only $386.15 over 1,000 trades yields an expectancy (E) of **$0.386 per trade**. In live execution environments, clearing fees, slippage, and exchange round-trip commissions will render this system highly net-negative.**
***   **Stop-Loss Ghosting:** The current `atrStopMultiplier` of **1164.26** is absurdly wide. This indicates that the stop-loss is functionally infinite. Risk is not being managed by price action; instead, trades are surviving due to high mean-reversion tendencies and are being terminated almost exclusively by the time-stop parameter (`maxTicksInTrade: 120`).**
***   **Drawdown Sub-Utilization:** A maximum peak drawdown of **0.4%** alongside a 1,000-trade sample suggests extreme capital under-allocation. The system is operating in a risk vacuum, suffocating potential absolute returns.**

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2. REGIME SUITABILITY & BEHAVIORAL PATTERNS

The configuration profile describes an aggressive mean-reversion engine bolted onto a trend-filtering framework that is currently misaligned.

Regime Conflict (ADX vs. RSI/BB):

The system uses `regimeAdxThreshold: 25`. An ADX > 25 defines a strong trending environment. Concurrently, the system uses Bollinger Bands (`bbStd: 2.25`) and RSI thresholds (`25 / 75`) to trade mean-reversion.
Behavioral Failure: The system is actively hunting for mean-reversion tops/bottoms inside strong trending markets (ADX > 25). This leads to "catching falling knives" behaviors. The high win rate is only preserved because the `maxTicksInTrade` time-exit bail-out occurs before the trend can fully liquidate the unstopped position.
Time-Decay Dependency: The system is highly dependent on short-term micro-oscillations. It does not capture macroeconomic or structural trends; it exploits noise.

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3. CALIBRATED RECOMMENDATIONS

To transform this system from a theoretical micro-scalper into an institutionally viable alpha generator, execute the following parameters sweeps immediately:

Parameter Re-Calibration Matrix

Parameter	Current Value	Optimized Value	Strategic Justification
`atrStopMultiplier`	1164.26	2.75 - 3.25	Hardens systemic risk. Stops "holding and hoping" behavior. Translates infinite tail-risk into a defined, predictable loss distribution.
`regimeAdxThreshold`	25	< 18 (Low Trend)	Restricts the mean-reversion engine to low-volatility, ranging markets where RSI and BB boundaries are highly respected.
`rsiOversoldThreshold`	25	30	Increases trade frequency in ranging regimes without sacrificing entry quality.
`rsiOverboughtThreshold`	75	70	Symmetrical normalization to align with standard deviation boundaries.
`bbStd`	2.25	2.00	Increases the sensitivity of the outer bands to capture high-probability touchpoints within low-ADX cycles.
`maxTicksInTrade`	120	85	Tightens the time-cap. Since the stop-loss is now active, we reduce capital lock-up duration in stale trades.

 # Risk & Leverage Scaling Directive:
1. **Introduce Sizing Escalation: Since Max Drawdown is exceptionally low at $0.4\% $, scale the contract/lot size by a factor of **5x to 8x**. This will scale the average trade net return above the commission threshold while keeping projected system drawdown well below a standard institutional limit of 5.0%.**
2. **Slippage Mitigation Filter: Implement a minimum tick-size filter. Do not execute trade entries unless the average bid-ask spread is $< 10\%$ of the projected average trade net expectancy ($E \approx \$0.38$).**
